Hints: Continuous functions – 2

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. The range is not an interval, so such an f cannot be continuous (Theorem 2). Look at a discontinuity point x_0.

Hint 2. At x_0 there is a jump: $f(x_0^-) < f(x_0^+)$. For $x < x_0$, $f(x) \leqslant f(x_0^-)$, and for $x > x_0$, $f(x) \geqslant f(x_0^+)$. So no value in $(f(x_0^-), f(x_0^+))$ other than $f(x_0)$ is taken, yet this interval contains many irrationals.

Problem 2

Hint 1. Answer: yes. It suffices to construct a strictly increasing map $(0,1) \to \mathbb{R} \setminus \mathbb{Q}$, since $\mathbb{R}$ is order-isomorphic to $(0,1)$. Think of decimal expansions.

Hint 2. Use the expansion $x = 0.d_1d_2d_3\ldots$ that does not end in 9's, and spread its digits out while inserting a fixed pattern that can never become periodic: $y = 0.d_1\,1\,d_2\,0\,1\,d_3\,0\,0\,1\,d_4\,0\,0\,0\,1\ldots$ (between d_k and d_{k+1} insert $k-1$ zeros and a 1).

Hint 3. y has blocks of zeros of every length, each followed by a 1, so its expansion is not eventually periodic and y is irrational. Neither expansion ends in all 0's or all 9's, so the lexicographic order of digits is the numerical order, and $x \mapsto y$ is strictly increasing.

Problem 3

Hint 1. Answer: exactly $\alpha = \frac{1}{n}$. For $\alpha = \frac{1}{n}$, consider $g(x) = f(x + \frac{1}{n}) - f(x)$ and sum g over $x = 0, \frac{1}{n}, \ldots, \frac{n-1}{n}$.

Hint 2. The sum telescopes to $f(1) - f(0) = 0$, so g is not of constant strict sign at these points; the IVT gives a zero.

Hint 3. For other α, try $f(x) = \sin^2 \frac{\pi x}{\alpha} - x \sin^2 \frac{\pi}{\alpha}$. Check that $f(0) = f(1) = 0$ and that $f(x+\alpha) - f(x) = -\alpha \sin^2 \frac{\pi}{\alpha} \neq 0$.

Problem 4

Hint 1. Consider $g(x) = f(x+T) - f(x)$ and suppose the claim fails. Then $|g| \geqslant \varepsilon$ on some half-line $[A, \infty)$.

Hint 2. By continuity g has a constant sign there, say $g \geqslant \varepsilon$. Then $f(A+nT) \geqslant f(A) + n\varepsilon \to \infty$, which contradicts boundedness.

Problem 5

Hint 1. For $n \geqslant 1$ consider $A_n = \{x : |f(x)| > \frac{1}{n}\}$. What does $\lim_{x \to x_0} f(x) = 0$ say about points of A_n near x_0?

Hint 2. A_n has no accumulation point, so it is finite in each bounded interval and hence countable. Therefore $\{f \neq 0\} = \bigcup_n A_n$ is countable, and f vanishes at all but countably many points.